

CONTACT INFORMATION

Date of Birth: June 14, 1998

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EDUCATION

California Institute of Technology

Sep. 2021 – May 2026

Ph.D. in Applied Physics (defended May 2026), advisor: Prof. Marco Bernardi

Milton and Francis Clauser Doctoral Prize (best thesis award across all disciplines.)

Nanjing University

Sep. 2016 – Jun. 2020

B.S. Physics, graduated with honors

RESEARCH INTERESTS

- Materials science, quantum many-body theory, computational physics
- Machine learning quantum interactions in materials
- Strong coupling, transport, electron-phonon interactions, anharmonic phonon

RESEARCH EXPERIENCE

• Postdoctoral Researcher

Department of Applied Physics and Materials Science

California Institute of Technology. Pasadena, CA, USA

Supervisor: Prof. Marco Bernardi

Dates: July 2026 – present

• Graduate Research Assistant

Department of Applied Physics and Materials Science, and Department of Physics

California Institute of Technology. Pasadena, CA, USA

Supervisor: Prof. Marco Bernardi

Dates: September 2021 – May 2026

Developed novel first-principles methods for quantum interactions in materials

• Research Assisant

Department of Physics

Nanjing University. Nanjing, Jiangsu, China

Supervisor: Prof. Qianghua Wang

Dates: September 2019 – July 2021

Learned quantum many-body theory and related simulation techniques

HONORS AND AWARDS

Milton and Francis Clauser Doctoral Prize	2026
(Awarded annually to a single Caltech Ph.D. graduate for the thesis of greatest originality.)	
Eddleman Graduate Fellowship	2024
B.S. in Nanjing University, Graduated with Honors	2020
National Scholarship, Ministry of Education of China	2018

TEACHING AND MENTORING

- **Teaching Assistant:** Spring 2026, Introduction to Computational Methods for Science and Engineering (MS 141), Caltech
- **Teaching Assistant:** Fall 2025, Structure and Bonding in Materials (MS 131), Caltech
- **Undergraduate Research Mentoring:** Supervised two Caltech SURF interns (Summer 2024)
 - *Dhruv Mangtani* — *Data-driven compression of phonon-phonon interactions*
 - *Alex Roger* — *Spin relaxation in the polaron region using diagrammatic quantum Monte Carlo*

SELECTED PUBLICATIONS

- [1] F. Liu^{*}, Z. Yang^{*}, **Y. Luo^{*}**, S. Guo, C. Zhang, S. Choo, X. Xu, S. G. Jeong, J. S. Kumar, X. Wang, A. Mkhoyan, M. Bernardi, and B. Jalan
Understanding polaronic transport in complex oxides by combining precise synthesis and first-principles many-body theory.
Rep. Prog. Phys. **2026** **89**, 028003
^{*}These authors contributed equally.
See the Research Highlight from Physics World: [How polarons travel through TiO₂](#).
- [2] **Y. Luo**, D. Mangtani, S. Peng, J. Yao, S. Kliavinek, and M. Bernardi
Tensor Learning and Compression of N-Phonon Interactions.
Phys. Rev. Lett. **2025** **135**, 126101
See the story about this paper on [Caltech News](#).
- [3] **Y. Luo**, J. Park, and M. Bernardi
First-principles diagrammatic Monte Carlo for electron–phonon interactions and polaron.
Nature Physics **2025** **21**, 1275–1282
Read the related news articles from [Caltech News](#) and [Physics World](#).
- [4] **Y. Luo**, D. Desai, B. K. Chang, J. Park, and M. Bernardi
Data-Driven Compression of Electron-Phonon Interactions.
Physical Review X **2024** **14**, 021023
See the story about this paper on [Caltech News](#).
- [5] **Y. Luo**, B. K. Chang, and M. Bernardi
Comparison of the canonical transformation and energy functional formalisms for ab initio calculations of self-localized polarons.
Physical Review B **2022** **105**, 155132

- [6] K. B. Le, A. Esquembre-Kucukalic, H.-Y. Chen, I. Maliyov, **Y. Luo**, J.-J. Zhou, D. Sangalli, A. Molina-Sánchez, and M. Bernardi
Magnon-phonon interactions from first principles.
Phys. Rev. B **2025 112**, L180403
- [7] S. Gao, J.-J. Zhou, **Y. Luo**, and M. Bernardi
First-principles electron-phonon interactions and electronic transport in large-angle twisted bilayer graphene.
Physical Review Material **2024 8**, L051001
- [8] J. Park, J.-J. Zhou, **Y. Luo**, and M. Bernardi
Predicting Phonon-Induced Spin Decoherence from First Principles: Colossal Spin Renormalization in Condensed Matter.
Physical Review Letter **2022 129**, 197201
See the story about this paper on [Caltech News](#).
- [9] J. Park, **Y. Luo**, J.-J. Zhou, and M. Bernardi
Many-body theory of phonon-induced spin relaxation and decoherence.
Physical Review B **2022 106**, 174404

TALKS

2026 The Penn Conference in Theoretical Chemistry. Philadelphia, PA.

Compressing Quantum Interactions in Matter: From Tensor Learning to Diagrammatics (postdoc award talk)

2025 American Physical Society March Meeting. Anaheim, CA.

Data-driven compression of first-principles phonon-phonon interactions

2024 American Physical Society March Meeting. Minneapolis, Minn.

Precise simulations of polarons in real materials: a first-principles diagrammatic Monte Carlo approach

2023 IQIM Postdoctoral and Graduate Student Seminar. Pasadena, CA.

Data-driven compression of electron-phonon interactions

2023 American Physical Society March Meeting. Las Vegas, NV.

Efficient compression of first-principles electron-phonon interactions

2022 American Physical Society March Meeting. Chicago, IL.

Comparing the canonical-transformation and energy-functional approaches for ab initio calculations of self-localized polarons

PROFESSIONAL ACTIVITIES

Reviewer: Physical Review X, Physical Review Letters, Physical Review B, The Journal of Chemical Physics

Open-source software: Contributor to the PERTURBO code for electron-phonon and transport calculations; presenter at two PERTURBO workshops.